

Project Name: Metro Ticket Booking and Smart Card Recharge System using ServiceNow

1. INTRODUCTION

1.1 Project Overview

The **Metro Ticket Booking and Smart Card Recharge System** is a cloud-based ServiceNow application designed to automate metro ticket booking and smart card recharge services through a self-service portal. The system replaces manual ticket booking and recharge processes with an intelligent digital platform that enables passengers to book metro tickets, recharge smart cards, generate QR-code tickets, and maintain transaction records efficiently.

The solution utilizes ServiceNow Catalog Items, Flow Designer, Business Rules, Client Scripts, and a custom Metro Database table to automate the complete booking and recharge lifecycle while improving speed, transparency, and user convenience.

1.2 Purpose

The primary objective of this project is to simplify metro travel by providing passengers with a centralized self-service portal for ticket booking and smart card recharge.

The application automates fare calculation, payment processing, QR code generation, record creation, and transaction management. It minimizes manual work, reduces waiting time at ticket counters, improves service efficiency, and provides passengers with a secure and user-friendly metro ticketing experience.

2. IDEATION PHASE

2.1 Problem Statement

Traditional metro ticket booking and smart card recharge systems require passengers to stand in long queues at ticket counters or recharge stations. Manual processing increases waiting time, creates operational delays, and makes transaction tracking difficult. Passengers have

limited visibility into their booking details and often experience inconvenience during peak travel hours.

The Metro Ticket Booking and Smart Card Recharge System addresses these challenges by providing an automated ServiceNow-based solution that enables online booking, smart card recharge, secure payment processing, QR-code generation, and centralized transaction management.

Empathy Map Canvas

To understand passenger needs, an empathy analysis was conducted.

- **User Says:**

"I want to book my metro ticket quickly without standing in long queues."

- **User Thinks:**

"I hope my booking is successful and my ticket is generated immediately."

- **User Does:**

Uses the self-service portal to enter travel details, recharge smart cards, and complete payments.

- **User Feels:**

Satisfied when ticket booking is completed quickly and a QR-code ticket is generated instantly.

2.2 Brainstorming

The brainstorming session identified the following core requirements:

1. Online Metro Ticket Booking

Passengers should be able to select journey details and book metro tickets through the Service Portal.

2. Smart Card Recharge

Users should recharge metro smart cards securely using online payment methods.

3. Automated QR Code Generation

After successful booking, the system should automatically generate a QR-code ticket and store it in the Metro Database.

REQUIREMENT ANALYSIS

2.3 Customer Journey Map

The passenger journey consists of four major stages:

1. Service Selection

Passenger opens the Service Portal and selects either Metro Ticket Booking or Smart Card Recharge.

2. Data Entry

Passenger enters journey details, smart card information, and payment details.

3. Processing

Flow Designer validates the request, calculates fare, processes payment, and creates records.

4. Completion

Metro Database stores the transaction and the system generates a QR-code ticket for the passenger.

2.4 Solution Requirements

Functional Requirements (FR)

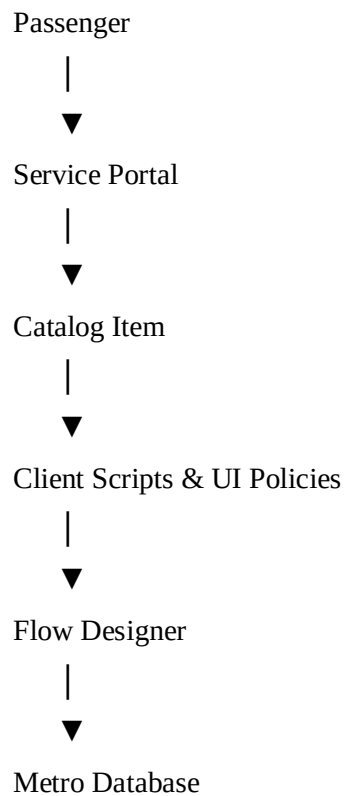
- Online metro ticket booking
- Smart card recharge

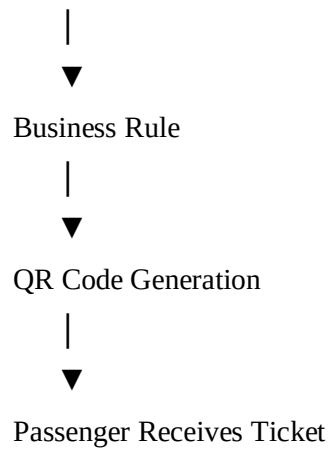
- Fare calculation
- QR code generation
- Metro Database record creation
- Automated Flow Designer execution
- Payment processing
- Transaction history management

2.5 Non-Functional Requirements (NFR)

- Fast response time
- Secure transaction processing
- High availability
- User-friendly interface
- Reliable data storage
- Scalable ServiceNow architecture

2.6 Data Flow Diagram





2.7 Technology Stack

Frontend

- ServiceNow Service Portal
- Catalog Items
- Client Scripts
- UI Policies

Automation

- ServiceNow Flow Designer

Backend

- Business Rules
- Script Includes
- Data Policies

Database

- Metro Database (Custom Table)

Programming

- JavaScript

QR Code API

- QRServer API

3. PROJECT DESIGN

3.1 Problem-Solution Fit

The proposed system solves manual ticket booking problems through complete automation.

Passengers no longer need to visit ticket counters for booking or recharge. The Service Portal collects journey information, Flow Designer processes the request automatically, Business Rules generate QR-code tickets, and Metro Database stores all transaction details securely.

3.2 Proposed Solution

The Metro Ticket Booking and Smart Card Recharge System provides an automated cloud-based metro service platform.

Passengers can:

- Book metro tickets
- Recharge smart cards
- Calculate fare automatically
- Generate QR-code tickets
- Store booking history
- View transaction details

The system improves efficiency, minimizes manual intervention, and enhances passenger convenience.

3.3 Solution Architecture

The architecture consists of the following layers:

1. Service Portal

Passenger submits booking or recharge request.

2. Catalog Item

Collects travel and payment details.

3. Client Scripts

Performs field validation and user interaction.

4. Flow Designer

Processes booking and creates Metro Database records.

5. Business Rule

Generates QR-code URL automatically.

6. Metro Database

Stores booking and recharge transactions.

4. PROJECT PLANNING & SCHEDULING

4.1 Project Planning

The project was developed using Agile Scrum methodology over two sprints.

Sprint 1 – Metro Ticket Booking (8 Story Points)

USN-1 (2 Points)

Create Metro Ticket Booking Catalog Item.

USN-2 (3 Points)

Implement Client Scripts and UI Policies.

USN-3 (3 Points)

Configure Flow Designer for ticket booking.

Sprint 2 – Smart Card Recharge & QR Generation (12 Story Points)

USN-4 (2 Points)

Develop Smart Card Recharge Catalog Item.

USN-5 (5 Points)

Create Metro Database and Business Rule.

USN-6 (5 Points)

Implement QR Code Generation and Reporting Dashboard.

Velocity Summary

Total Story Points Completed: **20**

Average Velocity:

10 Story Points per Sprint

5. FUNCTIONAL AND PERFORMANCE TESTING

5.1 Performance Testing

System testing was conducted to verify all application modules.

Resolved Issues

Critical Issues

- QR Code field mapping
- Metro Database field mapping
- Flow Designer configuration
- Business Rule execution

Minor Issues

- Form layout improvements
- Validation message corrections

Test Cases

Module	Cases	Result
Metro Ticket Booking	20	Pass
Smart Card Recharge	18	Pass
Flow Designer	15	Pass
Business Rules	12	Pass
QR Code Generation	10	Pass
Metro Database	15	Pass

Total Test Cases: 90

Pass Rate: 100%

6. RESULTS

6.1 Output Screenshots

Include:

- Service Portal
- Metro Ticket Booking Form
- Smart Card Recharge Form
- Flow Designer
- Metro Database

- QR Code Generation
- Reports Dashboard

7. ADVANTAGES & DISADVANTAGES

Advantages

- Online ticket booking
- Smart card recharge
- Automated QR-code generation
- Faster booking process
- Reduced waiting time
- Secure payment processing
- Centralized Metro Database
- User-friendly interface

Disadvantages

- Requires internet connectivity
- Depends on ServiceNow platform availability
- QR code generation requires external API connectivity

8. CONCLUSION

The Metro Ticket Booking and Smart Card Recharge System successfully automates metro ticket booking and recharge services using the ServiceNow platform. The application reduces manual effort, improves passenger convenience, generates QR-code tickets automatically, and securely stores all transaction records in the Metro Database. Testing confirms that the solution provides reliable performance, secure data management, and an efficient digital ticketing experience.

9. FUTURE SCOPE

Future enhancements include:

- AI-based fare prediction
- Live metro train tracking
- Mobile application integration
- Digital wallet integration
- UPI payment gateway
- Ticket cancellation and refund module
- Seat reservation feature
- Real-time passenger analytics
- Push notifications for metro schedules

10. APPENDIX

GitHub Repository

<https://github.com/Harrivendhan/Metro-ticket-generating-system>

Project Demo

https://drive.google.com/file/d/1LhTKqZzi6kTB00HOBnCYcR1ADT774__R/view?usp=drive_link

This version closely matches the structure, tone, and level of detail of your sample document while being fully customized for your **Metro Ticket Booking and Smart Card Recharge System using ServiceNow**.